Description

Image





Caption

1. Personal computer casing made of polycarbonate. © Chris Lefteri 2. Polycarbonate is tough and impact-resistant: hence its use in hard hats and helmets, transparent roofing and riot shields.

The material

PC is one of the 'engineering' thermoplastics, meaning that they have better mechanical properties than the cheaper 'commodity' polymers. The family includes the plastics polyamide (PA), polyoxymethylene (POM) and polytetrafluorethylene (PTFE). The benzene ring and the -OCOO- carbonate group combine in pure PC to give it its unique characteristics of optical transparency and good toughness and rigidity, even at relatively high temperatures. These properties make PC a good choice for applications such as compact disks, safety hard hats and housings for power tools. To enhance the properties of PC even further, it is possible to co-polymerize the molecule with other monomers (improves the flame retardancy, refractive index and resistance to softening), or to reinforce the PC with glass fibers (giving better mechanical properties at high temperatures).

Compositional summary

(O-(C6H4)-C(CH3)2-(C6H4)-CO)n

General properties

Density		1.19e3	-	1.21e3	kg/m^3
Price	*	2.2	-	2.53	GBP/kg
Date first used		1958			

Mechanical properties

Young's modulus		2.32	-	2.44	GPa
Shear modulus	*	0.829	-	0.872	GPa
Bulk modulus	*	3.83	-	4.03	GPa
Poisson's ratio	*	0.391	-	0.407	
Yield strength (elastic limit)		59.1	-	65.2	MPa

Tensile strength		62.7	-	72.4	MPa
Compressive strength	*	69	-	86.2	MPa
Elongation		110	-	150	% strain
Hardness - Vickers	*	18	-	20	HV
Fatigue strength at 10^7 cycles	*	23.7	-	30.8	MPa
Fracture toughness		2.1	-	2.3	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0164	-	0.0172	

Thermal properties

Glass temperature		142	-	158	°C
Maximum service temperature	*	101	-	116	°C
Minimum service temperature		-47.2	-	-37.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.193	-	0.218	W/m.°C
Specific heat capacity		1.15e3	-	1.25e3	J/kg.°C
Thermal expansion coefficient		120	-	125	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		1e20	-	1e21	µohm.cm
Dielectric constant (relative permittivity)		3.1	-	3.3	
Dissipation factor (dielectric loss tangent)		8.6e-4	-	9.4e-4	
Dielectric strength (dielectric breakdown)	*	16	-	19.2	MV/m

Optical properties

Transparency		Optical Quality			
Refractive index		1.58	-	1.59	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			

Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Limited use

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Unacceptable
Carbon tetrachloride	Excellent
Chloroform	Unacceptable
Crude oil	Limited use
Diesel oil	Limited use
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Unacceptable
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Unacceptable
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Limited use

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Acceptable
Marine atmosphere	Acceptable
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Slow-burning

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO_2-eq) (virgin grade production)	* 4.94	-	5.44	kg/kg
Embodied energy (virgin grade production)	* 109	-	120	MJ/kg
Water usage	* 165	-	182	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.06	-	8.93	MJ/kg
Polymer molding (injection, blow), embodied energy	* 21	-	24.4	MJ/kg
Machining, coarse, embodied energy	* 18	-	20.7	MJ/kg

Machining, fine, embodied energy	* 24.8	-	29.9	MJ/kg
Machining, grinding, embodied energy	* 31.5	-	39	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.53	-	0.589	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.46	-	1.71	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.18	-	1.36	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.62	-	1.96	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.06	-	2.56	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.63	-	1.8	kg/kg
Embodied energy, recycling	* 20.7	-	22.8	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 30.3	-	31.8	MJ/kg
Combustion CO2	* 2.7	-	2.84	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

The processing of engineering thermoplastics requires a higher energy input than that of commodity plastics, but otherwise there are no particular environmental penalties. PC can be recycled if unreinforced.

Recycle mark



Supporting information

Design guidelines

The optical transparency and high impact resistance of PC make it suitable for bullet-resistant or shatter-resistant glass applications. It is readily colored. PC is usually processed by extrusion or thermoforming (techniques that impose constraints on design), although injection molding is possible. When designing for extrusion with thermoplastics, the wall thickness should be as uniform as possible to prevent warping, and projections and sharp corners avoided- features like hollows and lone unsupported die sections greatly increase the mold cost. The stiffness of the final part can be improved by the incorporation of corrugations or embossed ribs. PC can be reinforced using glass fibers to reduce shrinkage problems on cooling and to improve the mechanical performance at high temperatures.

Technical notes

The combination of the benzene ring and carbonate structures in the PC molecular structure give the polymer its unique characteristics of high strength and outstanding toughness. It can be easily blended with ABS or polyurethane. ABS/PC gets its flame retardance and UV resistance from polycarbonate at a lower cost than that of ABS. PU/PC gets its rigidity from polycarbonate and flexibility and ease of coating from polyurethane.

Typical uses

Safety shields and goggles, lenses, glazing panels, business machine housing, instrument casings, lighting fittings, safety helmets, electrical switchgear, laminated sheet for bullet-proof glazing, twin-walled sheets for glazing, kitchenware and tableware, microwave cookware, medical (sterilizable) components.

Tradenames

Calibre, FR-PC, Latilon, Lexan, Lupilon, Makrolon, Naxell, Nyloy, Panlite, Sinvet, Star-C, Starglas, Triex, Xantar

Links

ProcessUniverse

Producers

Reference

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